

RACIBORZ, Poland

WMO: 125400

Lat: 50.0610N

Lon: 18.1908E

Elev: 676

StdP: 14.34

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.4	11.0	0.8	5.9	5.9	6.2	7.8	12.4	26.1	36.2	23.9	36.1	4.4	150	0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.7	86.8	68.4	83.2	67.1	80.2	65.8	70.4	81.9	68.7	79.7	67.3	77.3	8.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.6	100.3	75.0	65.1	95.0	73.3	63.6	90.3	71.7	34.7	81.8	33.3	79.7	32.1	77.6	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.2	19.1	16.7	0.7	92.4	8.1	3.1	-5.2	94.7	-9.9	96.5	-14.4	98.2	-20.3	100.5		
(5)			-0.5	73.9	8.1	1.7	-6.4	75.1	-11.1	76.1	-15.7	77.1	-21.6	78.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.8	29.2	32.1	39.1	49.4	57.0	63.1	66.6	66.2	57.9	49.8	41.5	33.1
	DBStd	14.98	9.50	9.14	8.02	7.32	6.57	6.18	5.68	5.83	6.14	7.28	7.30	8.70
	HDD50	2497	646	502	345	98	14	0	0	0	7	97	265	523
	HDD65	6164	1111	922	804	469	258	109	49	54	224	472	704	988
	CDD50	2073	0	0	6	79	232	394	514	502	245	90	11	0
	CDD65	265	0	0	0	1	10	52	99	91	11	1	0	0
	CDH74	2663	0	0	0	28	146	504	990	873	119	3	0	0
	CDH80	740	0	0	0	2	18	124	306	270	20	0	0	0
Wind	WSAvg	7.3	8.4	8.2	8.4	7.5	6.9	6.4	6.3	5.8	6.2	6.8	7.5	8.5
Precipitation	PrecAvg	24.8	1.3	1.1	1.4	1.6	2.9	3.1	3.7	2.7	2.4	1.7	1.7	1.3
	PrecMax	36.0	2.4	2.7	3.1	3.0	9.2	5.8	12.4	5.6	5.3	4.1	3.6	3.2
	PrecMin	13.7	0.4	0.2	0.5	0.3	0.7	0.6	0.4	0.5	0.6	0.2	0.0	0.4
	PrecStd	4.5	0.5	0.6	0.7	0.7	1.6	1.3	2.6	1.3	1.3	1.1	0.8	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.6	54.2	64.7	77.8	82.3	88.5	90.9	91.7	82.9	73.2	62.9	53.8
		MCWB	46.9	47.5	51.6	60.3	64.1	70.1	69.1	68.8	64.7	61.1	55.1	48.8
	2%	DB	46.7	49.8	59.0	71.6	78.0	83.6	87.0	86.6	77.2	68.7	58.2	49.5
		MCWB	43.5	44.7	49.6	57.0	63.0	68.1	68.3	68.1	63.3	59.3	52.3	46.1
	5%	DB	43.7	46.7	54.7	67.0	74.4	79.7	83.2	82.3	73.2	64.8	55.1	46.2
		MCWB	41.0	42.6	47.4	54.4	61.6	66.1	67.2	67.1	61.8	57.5	50.4	43.3
	10%	DB	41.0	43.5	50.8	63.0	70.5	75.9	79.7	78.7	69.6	61.0	51.9	43.4
		MCWB	38.6	40.3	45.3	52.6	59.5	64.6	65.9	65.6	60.5	55.5	48.5	41.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.6	48.1	53.1	61.5	67.3	72.1	72.6	72.6	66.9	62.7	55.4	49.2
		MCDB	50.3	53.6	62.3	74.1	78.4	84.4	86.0	83.6	78.0	70.8	61.6	53.0
	2%	WB	43.8	45.2	50.4	58.5	64.6	69.8	70.3	69.7	64.8	60.1	52.8	46.2
		MCDB	46.6	49.3	57.7	70.1	74.6	80.9	81.6	82.0	73.1	66.5	57.7	49.4
	5%	WB	41.1	42.9	48.0	55.8	62.7	67.6	68.8	68.1	63.1	58.0	50.7	43.5
		MCDB	43.5	46.3	53.6	64.9	72.3	76.8	79.5	79.5	71.1	63.6	54.4	46.0
	10%	WB	38.8	40.4	45.6	53.5	60.6	65.6	67.2	66.6	61.4	56.0	48.7	41.2
		MCDB	40.8	43.3	50.3	61.4	69.0	73.8	77.0	76.6	68.4	60.8	51.8	43.3
Mean Daily Temperature Range	5% DB	MDBR	8.9	10.4	13.6	18.3	18.3	18.2	19.7	19.7	16.8	14.5	10.1	8.5
		MCDBR	9.8	14.5	21.8	25.9	25.6	25.2	26.8	26.3	24.0	20.7	15.1	11.1
		MCWBR	7.9	10.4	13.5	13.5	12.6	12.0	10.6	10.5	11.6	12.4	10.6	8.7
	5% WB	MCDBR	9.5	13.9	19.6	23.6	22.8	22.2	23.4	23.1	20.8	18.7	14.1	10.8
MCWBR		7.8	10.4	12.7	13.0	12.1	11.6	10.4	10.0	10.8	11.6	10.3	8.7	
Clear-Sky Solar Irradiance	taub	0.316	0.341	0.381	0.434	0.429	0.429	0.444	0.435	0.404	0.373	0.346	0.312	
	taud	2.306	2.280	2.230	2.133	2.191	2.230	2.195	2.229	2.286	2.342	2.362	2.363	
	Ebn at Noon	225	249	258	256	263	264	257	253	248	234	212	210	
	Edh at Noon	22	29	36	44	44	43	43	40	34	27	21	18	
All-Sky Solar Radiation	RadAvg	268	503	851	1293	1562	1660	1593	1434	1017	605	299	207	
	RadStd	47	80	135	161	193	166	190	128	149	88	41	25	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (50 neighbors)		+0.90	N/A	N/A	+1.35	+0.67	+0.59	N/A	-277	+113	+40

Nomenclature: See separate page